Subseasonal Forecast Skill over India: A Two-Season Benchmark of AI-Based and Operational Models

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Abstract

Subseasonal-to-seasonal (S2S) forecasts at two-to-six-week leads are of high value for Indian agriculture and water management, yet precipitation skill at these leads remains limited, and the performance of the new generation of machine-learning (ML) forecast systems over India is largely undocumented. We benchmark six S2S systems—three ML-based (Spire AI-S2S, FuXi-S2S, DLESyM) and three operational dynamical systems (ECMWF, UKMO, NCEP)—for weekly precipitation, 500-hPa geopotential height (Z500), and 2-m temperature (T2M) over India across two physically contrasting seasons: the synoptically driven winter (JFM 2026, 90 daily initializations) and the convectively dominated summer monsoon (JJAS 2019, 17 common initializations). Forecasts are verified against ERA5 and IMD gridded rainfall using cosine-weighted anomaly correlation (ACC), RMSE, bias, and a common probabilistic framework (CRPSS, spread–skill ratio). Four findings emerge. (i) In winter, Spire AI-S2S is the most skillful system for precipitation at every lead (week-1 ACC 0.78 versus 0.73 for ECMWF and UKMO) and is the only system that retains Z500 skill to week 6 (ACC 0.75), while the other two ML systems—FuXi-S2S and DLESyM—collapse to negative Z500 ACC by week 3–4. (ii) In the monsoon, this ML advantage disappears entirely: every system, ML and dynamical alike, loses precipitation skill by week 3 (all-India ACC ≤ 0.06), and the result is unchanged on an independent 35-initialization robustness check. (iii) The three ML systems do not behave alike: Spire's long-lead stability is not shared by FuXi-S2S or DLESyM, so a week-1 leaderboard is a poor predictor of week-6 performance. (iv) Deterministic and probabilistic rankings disagree— ECMWF leads precipitation CRPSS from week 2 onward despite Spire's higher ACC—showing that ensemble calibration is a distinct axis of forecast quality. Together these results give an early, honest assessment of where ML S2S systems help over India and where the forecasting problem remains hard for every approach.

Keywords: subseasonal-to-seasonal prediction; machine learning; precipitation; India; monsoon; ensemble verification

1 Introduction

Subseasonal-to-seasonal (S2S) prediction, spanning lead times from roughly two weeks to two months, bridges medium-range weather forecasting and seasonal climate outlooks (Vitart et al., 2017; National Academies, 2016). Forecasts at these leads are increasingly demanded for agricultural planning, reservoir operations, energy load management, and disaster preparedness (White et al., 2017), yet they remain among the most difficult products to produce skillfully: the atmosphere has largely lost its initial-condition memory, while slowly varying boundary forcings have not yet come to dominate (Mariotti et al., 2020). Over India, skillful multi-week outlooks carry particular weight. Winter precipitation over northern India and the Western Himalaya, produced chiefly by western disturbances (WDs)—extratropical synoptic systems embedded in the subtropical westerly jet (Dimri et al., 2015; Hunt et al., 2018)—sustains the rabi (winter) cropping season and builds the Himalayan snowpack that feeds the Indus and Ganges during the subsequent melt season. The summer monsoon (June through September), by contrast, supplies roughly three-quarters of India's annual rainfall through transient convective systems organized by intraseasonal oscillations, and its subseasonal predictability is a long- standing operational priority for the India Meteorological Department (IMD) and the Ministry of Earth Sciences (Rao et al., 2019; Pattanaik et al., 2019). These two seasons are governed by different dynamics and, as we show below, by different limits on predictability.

In parallel, weather prediction is being reshaped by machine-learning (ML) forecast systems trained on reanalysis data. Deterministic data-driven models such as Pangu-Weather and GraphCast now match or exceed operational numerical weather prediction (NWP) on standard medium-range scores (Pathak et al., 2022; Bi et al., 2023; Lam et al., 2023; Ben Bouallègue et al., 2024; Rasp et al., 2024), and operational centers have begun running data-driven systems alongside their NWP suites (Lang et al., 2024). Extension of these methods to the subseasonal range is more recent: FuXi-S2S, a transformer-based system producing 42-day ensemble forecasts, has been reported to outperform conventional S2S models on several global metrics (Chen et al., 2024); diffusion-based Earth-system emulators such as DLESyM (Cresswell-Clay et al., 2025) have followed; and commercial providers now issue AI-based S2S products (Spire Global, 2025). Published evaluations of these systems have, however, emphasized global or large-scale diagnostics, while regional precipitation skill over densely populated, hydrologically sensitive domains such as India remains largely undocumented—and essentially no published work compares more than one ML S2S system against operational dynamical baselines over the same domain and the same verification framework.

A further gap concerns the evaluation setting itself. Existing ML weather-forecast evaluations are typically run on a single season or a short real-time window, which cannot distinguish a genuinely better forecast system from one that merely performs well in an easy predictability regime. India offers an unusually clean natural test of this distinction: its two principal rainy seasons are governed by essentially different physics. Winter precipitation is organized by synoptic-scale WDs with several days of predictable structure, while monsoon precipitation is dominated by smaller- scale convection embedded in a noisier large-scale envelope, historically associated with a shorter predictability horizon (de Andrade et al., 2019; Pegion et al., 2019). A forecast system that performs well only in the synoptically organized winter season may not be skillful in general; a system that holds skill in the monsoon, where the signal-to-noise ratio is lower, is making a stronger claim. We exploit this contrast deliberately, evaluating the same six S2S systems—three ML-based and three operational dynamical—over India in both the

January–March (JFM) 2026 winter and the June–September (JJAS) 2019 summer monsoon, using an identical verification framework in both seasons.

Specifically, we address four questions:

1. How does weekly skill for precipitation, Z500, and T2M decay with lead time (weeks 1–6) for each system, in winter and in the monsoon?
2. Do ML systems outperform operational dynamical systems over India, and does any advantage persist across both seasons?
3. Do the ML systems behave consistently with one another, or do they diverge at long lead?
4. Does deterministic skill translate into probabilistic (calibrated) skill?

We verify ensemble-mean (or Spire mean) forecasts against ERA5 and IMD gridded rainfall using the cosine-weighted anomaly correlation coefficient (ACC), root-mean-square error (RMSE), and bias, complemented by a common-framework probabilistic evaluation (CRPSS, spread–skill ratio). The remainder of the paper is organized as follows. Section 2 describes the verification data and study domain, Section 3 the six forecast systems, and Section 4 the verification methodology. Results are presented in Section 5 and discussed in Section 6; limitations and conclusions follow in Sections 7–8.

2 Data and Study Domain

2.1 Study domain and regions

The study domain covers the Indian region, 5°–38°N, 65°–100°E, with all aggregate skill metrics computed over *land points only.* Land points are identified on the common 1.5° verification grid with an offline coastline land–sea mask, independent of ERA5 and fully reproducible; ocean grid cells (Arabian Sea, Bay of Bengal) are excluded from every area average, since they otherwise dominate the domain mean and distort precipitation skill. Spatial means are weighted by the cosine of latitude throughout (Section 4.3). Skill is additionally partitioned into the four India Meteorological Department (IMD) homogeneous rainfall regions—Northwest, Central, South Peninsula, and East/Northeast India—following the state grouping of Pai et al. (2014). The region masks are constructed from Survey-of-India state-boundary polygons by assigning each grid point to the region whose constituent states contain it, then intersecting with the land mask; the four regions are mutually exclusive.

2.2 Verification references

Three observational references are used, chosen to match the best available truth in each season and to avoid conflating model skill with reference-data artifacts.

- **Precipitation, JFM 2026:** hourly total precipitation from the Analysis-Ready, Cloud-Optimized ERA5 archive (ARCO-ERA5; Carver and Merose, 2023), summed to true 24-hour daily totals (00–00 UTC). The 2026 IMD gridded rainfall product is not yet available for this season, so ERA5 is the only complete reference; we discuss the implications of treating a reanalysis as truth in Section 7.

Table 1: Forecast systems evaluated. "Native res." is each model's own horizontal grid; all are regridded to the common 1.5° verification grid before scoring. CF = control, PF = perturbed member. "Both" = evaluated in JFM 2026 and JJAS 2019.

System	Type	Native res.	Members	Horizon	Seasons
Spire AI-S2S	ML (data-driven)	0.5°	mean+s.d.	46 d	JFM 2026
FuXi-S2S	ML (transformer)	1.5°	1CF+10PF	42 d	Both
DLESyM	ML (diffusion)	0.25°	80 PF	48 d	Both
ECMWF	Dynamical, coupled	Tco319 (∼36 km)	100 PF	46 d	Both
UKMO GloSea6	Dynamical, coupled	N216 (∼60 km)	1CF+3PF	60 d	Both
NCEP CFSv2	Dynamical, coupled	T126 (∼100 km)	15 PF	44 d	Both

- **Precipitation, JJAS 2019:** the IMD 0.25° daily gauge- based gridded rainfall product (Pai et al., 2014), constructed from roughly 3000 quality-controlled rain-gauge stations by ordinary kriging. Because this reference is independent of any reanalysis, it removes the possibility that an ERA5-trained ML system is rewarded simply for reproducing ERA5's own precipitation biases.
- **Z500 and T2M, both seasons:** ERA5 (Hersbach et al., 2020). For JFM 2026, Z500 truth is taken from ERA5 CDS daily files; for JJAS 2019 we use the local WeatherBench2 ERA5 archive (Rasp et al., 2024), which provides matching geopotential and 2-m temperature fields at 1.0° resolution for the 2019 period.

Because the JFM and JJAS precipitation truths differ (ERA5 versus IMD), all within-season model rankings are valid, but the absolute precipitation skill level should not be compared directly across seasons; we return to this point in Section 7.

2.3 Forecast systems

We evaluate six S2S forecast systems, summarized in Table 1: three machine-learning systems (Spire AI-S2S, FuXi-S2S, DLESyM) and three operational dynamical systems (ECMWF extended range, UKMO GloSea6, NCEP CFSv2). Five systems—all but Spire, for which 2019 data were not available—are evaluated in both seasons, giving a fully matched six-system comparison in JFM 2026 and a five-system comparison in JJAS 2019. Full system descriptions, including how each variable is delivered and any unit conversions applied before scoring, are given in Section 3.

3 Forecast Systems

We evaluate six S2S forecast systems spanning three machine-learning systems and three dynamical NWP ensembles. The dynamical systems (ECMWF, UKMO, NCEP) are obtained from the WWRP/WCRP S2S database (Vitart et al., 2017); the ML systems were received directly from their respective providers. All six systems are evaluated over JFM 2026; five (excluding Spire) participate in the JJAS 2019 case study.

3.1 Spire AI-S2S

Spire AI-S2S (Spire Global, 2025) is a commercially produced, ML-based subseasonal forecasting system developed by Spire Global. Forecasts are delivered on a 0.5° global grid at daily temporal resolution to a 46-day horizon, initialized from 00 UTC ERA5 analyses. Rather than distributing individual ensemble members, the system supplies ensemble summary products—the mean and standard deviation, used here for both deterministic and probabilistic verification. Spire delivers daily total precipitation ($\mathrm{mm\,day^{-1}}$) directly, so no unit conversion is required. Forecasts were received for the JFM 2026 season under a research data-use agreement; 2019 data were not available, so Spire is excluded from the JJAS 2019 comparison.

3.2 FuXi-S2S

FuXi-S2S (Chen et al., 2024) is a transformer-based ML subseasonal forecasting system trained on ERA5 reanalysis. The published system generates 42-day global forecasts on a 1.5° grid as an 11-member ensemble (1 control forecast, 10 perturbed forecasts), initialized from 00 UTC ERA5 fields; we use the ensemble mean for deterministic scores and the member spread for probabilistic verification. The FuXi-S2S precipitation channel is archived as a mean rate ($\mathrm{mm\,h^{-1}}$) rather than a daily total and is multiplied by 24 before scoring to convert to $\mathrm{mm\,day^{-1}}$ (Section 4.3).

3.3 DLESyM

DLESyM (Deep Learning Earth System Model; Cresswell-Clay et al. 2025) is a diffusion-based coupled atmosphere–land–ocean ML system trained on ERA5 and CMIP6 simulations. Forecasts run on a 0.25° global grid to a 48-day horizon and are delivered as an 80-member ensemble—the largest ensemble in this comparison—which is particularly advantageous for probabilistic verification. DLESyM outputs 6-hourly fields, which we average to daily means before aggregating to weekly lead windows; geopotential is converted to geopotential height by dividing by $g = 9.80665\,\mathrm{m\,s^{-2}}$. DLESyM does not currently provide a precipitation channel and is therefore evaluated for Z500 and T2M only. Three per-initialization forecast stores were found to be corrupted (one reading as all-zero, two missing entirely) and are excluded from the comparable-initialization sets; an automated guard rejects any store that reads as all-zero at the first lead.

3.4 ECMWF extended-range ensemble

The ECMWF Integrated Forecasting System (IFS) extended-range ensemble (Vitart, 2014) is a fully coupled atmosphere–ocean–land–sea-ice model run twice weekly to day 46. We use the 00 UTC runs archived on the S2S database (Vitart et al., 2017) at 1.5° (interpolated from the native Tco319, ≈36 km, spectral grid), with 100 perturbed-forecast members per initialization; we use the ensemble mean for deterministic scores and the full ensemble for probabilistic verification. ECMWF archives total precipitation as a running cumulative accumulation from initialization; the weekly-mean rate is recovered by differencing the cumulative totals at the start and end of each lead window and dividing by the number of days (Section 4.3). Geopotential height is delivered directly and requires no unit conversion.

3.5 UKMO GloSea6

The UK Met Office Global Seasonal Forecasting System version 6 (GloSea6; MacLachlan et al., 2015) is a fully coupled atmosphere–ocean–land system run on the Met Office Unified Model (MetUM) at ~60 km (N216) resolution to a 60-day horizon. We use the 00 UTC runs archived on the S2S database at 1.5°, comprising one control forecast plus three perturbed members (a four-member ensemble), which limits UKMO's probabilistic scores relative to the larger ensembles but still supports full deterministic verification. UKMO delivers total precipitation as a cumulative accumulation, processed identically to ECMWF, and geopotential height already in gpm (no unit conversion). The UKMO 2-m temperature archive contains only a single forecast step in every file we obtained, across all seasons; we treat this as a data-retrieval issue at the source and exclude UKMO T2M from all verification and rankings.

3.6 NCEP CFSv2

The NCEP Climate Forecast System version 2 (Saha et al., 2014) is a fully coupled atmosphere–ocean–land system, operational since 2011, resolved at T126 (≈100 km) with 64 hybrid sigma-pressure levels. Forecasts are archived on the S2S database at 1.5° to a 44-day horizon; we use the 15-member perturbed-forecast ensemble from each 00 UTC initialization. Precipitation is archived as a cumulative accumulation and de-accumulated identically to ECMWF and UKMO.

4 Verification Framework

4.1 Weekly lead windows and aggregation

Each forecast is partitioned into six nonoverlapping weekly lead windows—week 1 to forecast days 1–7, week 2 to days 8–14, and so on through week 6 (days 36–42), where day 1 denotes the first 24 h after initialization. For each lead window, daily fields are averaged to a weekly-mean field. All forecasts and the observational references are remapped to a common 1.5° latitude–longitude grid (matching the coarsest delivered system) prior to verification, so that no high-resolution system is penalized for representing fine scales the others cannot. Spatial means use $\cos\varphi$ area weights and are restricted to Indian land points (Section 2.1).

Because the six systems deliver precipitation in different native forms, each is reconciled to a common weekly-mean rate (mm day^{-1}) before scoring. Spire and FuXi-S2S (after the ×24 unit conversion, Section 3.2) deliver daily values that are simply averaged over the seven days of the lead window. ECMWF, UKMO, and NCEP archive precipitation as a running cumulative accumulation from initialization; the weekly-mean rate is recovered by differencing the cumulative totals at the last day (d_e) and the day before the window opens ($d_s - 1$), then dividing by the number of days N:

$$\overline{P}_w = \frac{\mathrm{TP}(d_e) - \mathrm{TP}(d_s - 1)}{N}, \tag{1}$$

with $\mathrm{TP}(d_s - 1) = 0$ for week 1. The ERA5 and IMD truth references are constructed from true 24-hour daily totals and then averaged over the same lead window.

4.2 Anomalies and climatology

Both forecast and observed anomalies are defined by removing a single common climatology, applied identically to every system and to the truth: a 30-year (1990–2019) ERA5 day-of-year climatology for Z500, T2M, and ERA5-referenced precipitation, and the IMD 1991–2020 daily rainfall climatology (Pai et al., 2014) for IMD-referenced precipitation. Because each climatology is computed over decades independent of the verification season, it avoids the amplitude inflation that a same-season baseline would introduce.

4.3 Deterministic and probabilistic metrics

The anomaly correlation coefficient (ACC) at lead ℓ is the cosine-latitude-weighted, spatially centered Pearson correlation of forecast and observed anomaly fields, computed per initialization and averaged over the season:

$$\mathrm{ACC}(\ell) = \frac{1}{T}\sum_{t=1}^{T} \frac{\sum_i w_i\,(f'_{i,t,\ell} - \overline{f'}_{t,\ell})\,(o'_{i,t,\ell} - \overline{o'}_{t,\ell})}{\sqrt{\sum_i w_i\,(f'_{i,t,\ell} - \overline{f'}_{t,\ell})^2}\,\sqrt{\sum_i w_i\,(o'_{i,t,\ell} - \overline{o'}_{t,\ell})^2}}\,, \tag{2}$$

where $w_i = \cos\varphi_i$ and $\overline{x}_{t,\ell}$ is the cosine-weighted spatial mean over land points. RMSE and bias are computed on the weekly-mean fields in their native units (mm day^{-1} for precipitation, m for Z500, K for T2M):

$$\mathrm{RMSE}(\ell) = \frac{1}{T}\sum_{t=1}^{T}\sqrt{\frac{\sum_i w_i\,(f_{i,t,\ell} - o_{i,t,\ell})^2}{\sum_i w_i}}\,, \qquad \mathrm{Bias}(\ell) = \frac{1}{T}\sum_{t=1}^{T}\frac{\sum_i w_i\,(f_{i,t,\ell} - o_{i,t,\ell})}{\sum_i w_i}\,. \tag{3}$$

For probabilistic verification, each system is placed on a common footing by treating it as a Gaussian forecast with ensemble mean μ and spread (standard deviation) σ at each grid point and lead day, regardless of whether the underlying product is individual members (FuXi-S2S, ECMWF, UKMO, NCEP, DLESyM) or a delivered mean-and-spread product (Spire). The continuous ranked probability score (CRPS) then has the closed form

$$\mathrm{CRPS}(\mu,\sigma,y) = \sigma\Big[\omega\,(2\Phi(\omega) - 1) + 2\phi(\omega) - \tfrac{1}{\sqrt{\pi}}\Big]\,, \quad \omega = \tfrac{y-\mu}{\sigma}, \tag{4}$$

with y the observation and Φ, ϕ the standard-normal CDF and PDF (Gneiting and Raftery, 2007). The skill score $\mathrm{CRPSS} = 1 - \mathrm{CRPS}/\mathrm{CRPS}_{\mathrm{clim}}$ is taken against a Gaussian climatological reference. Calibration is summarized by the spread–skill ratio $\mathrm{SSR} = \overline{\sigma}/\mathrm{RMSE}(\mu)$, which equals one for a perfectly reliable ensemble and is below one when the ensemble is overconfident (under-dispersive). Brier scores for precipitation exceedance events use the Gaussian exceedance probability $P(y > \tau) = 1 - \Phi\left(\frac{\tau-\mu}{\sigma}\right)$ for thresholds $\tau = 1$ and 10 mm day^{-1}.

4.4 Initialization sets and matched comparison

The two seasons use different initialization protocols, reflecting both data availability and the goal of a fair, common-init comparison within each season.

- **JFM 2026:** 90 daily 00 UTC initializations (1 January–31 March 2026) for the six-system comparison. Because the ERA5 daily truth record used for JFM currently ends

31 March 2026, the sample of initializations contributing to the longer lead windows shrinks at weeks 5–6 (the latest initializations cannot yet be verified at day 35–42); this does not bias the reported scores but reduces their precision at the longest leads, and we treat week-5–6 results accordingly.

- **JJAS 2019:** 17 common Thursday initializations (6 June–26 September 2019) on which all five non-Spire models provide usable forecasts; this matched set is the basis of all cross-model JJAS comparisons in Section 5.3. A separate 35-initialization operational subset (ECMWF, UKMO, NCEP only, twice-weekly Monday/Thursday) is used as a robustness check on the precipitation result (Section 5.7).

5 Results

5.1 Overview: skill across variables and seasons

Figure 1 summarizes the whole benchmark: anomaly correlation versus lead week for all six systems, three variables, and both seasons. Three features stand out immediately. First, in the top row (JFM 2026), the precipitation and T2M panels show Spire AI-S2S (blue) sitting visibly above every other system at every lead, while in the Z500 panel Spire is the only curve that does not bend downward after week 2—FuXi-S2S (orange) and DLESyM (green) instead fall through ACC= 0 by week 3–4. Second, the bottom row (JJAS 2019) looks qualitatively different: all six precipitation and Z500 curves converge toward, and then below, ACC= 0 by week 3, with no system maintaining the winter-season separation between ML and dynamical systems. Third, the two AI systems that are evaluated in both seasons, FuXi-S2S and DLESyM, show the steepest skill loss of any system in *either* season for Z500—a pattern we examine directly in Section 5.4. The following subsections quantify each of these features in turn.

5.2 Winter (JFM 2026): the ML advantage

Precipitation. Spire AI-S2S has the highest precipitation ACC of all six systems at every lead week (Table **??**): 0.78 at week 1, versus 0.73 for both ECMWF and UKMO and 0.62 for NCEP, narrowing to 0.31 by week 6 where it remains tied with the multi-model ensemble (MME) for the top score. Spire also has the lowest RMSE at every lead (Table **??**; 1.09 mm day^{-1} at week 1, rising to 2.28 mm day^{-1} at week 6, against 1.27–1.57 and 2.47–2.72 mm day^{-1} respectively for the other individual systems), so the ACC advantage is not an artifact of a smoother forecast field receiving a more favorable correlation score. FuXi-S2S, after the unit harmonization of Section 3.2, is competitive with the operational dynamical systems (ACC 0.72 at week 1) but does not match Spire at any lead.

Z500. The clearest separation in the entire benchmark appears in Z500 (Table **??**). At week 1 the field is tightly clustered (ACC 0.90–0.97 across all six systems, with ECMWF and the MME marginally ahead at 0.97), and all three ML systems are essentially indistinguishable from the dynamical systems. By week 3 the field has split sharply: Spire, ECMWF, UKMO, and NCEP all remain skillful (ACC 0.51–0.69), while FuXi-S2S has fallen to 0.05 and DLESyM

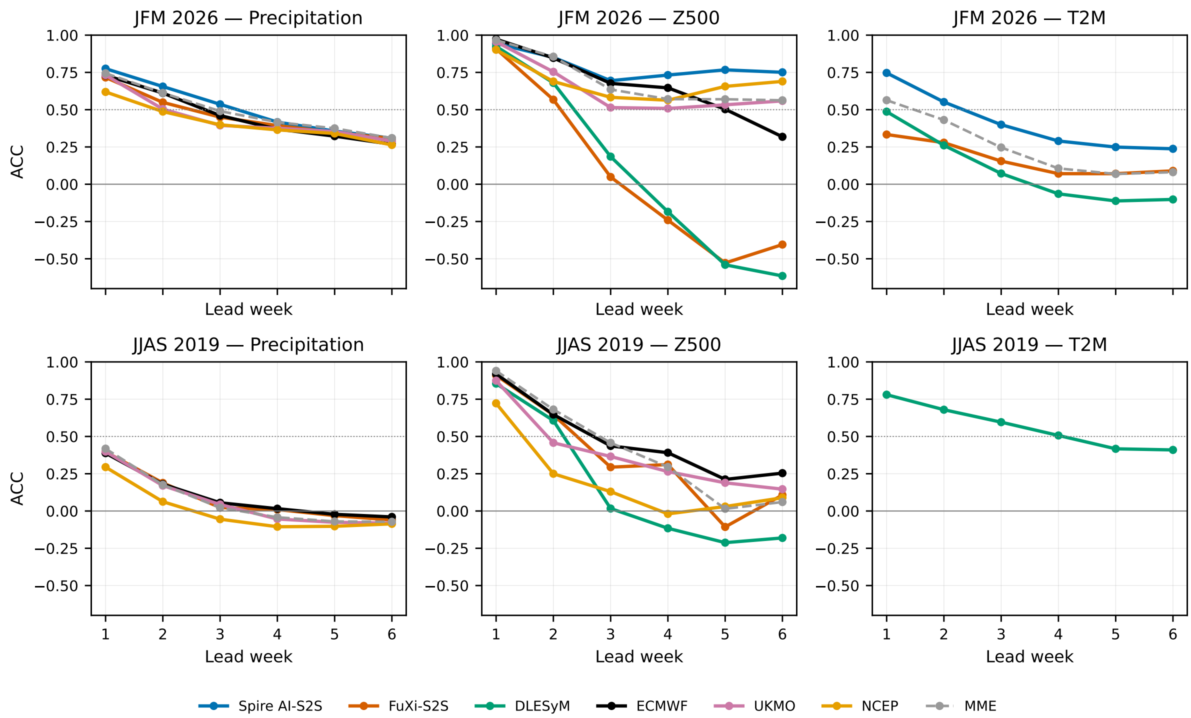

Figure 1: All-India anomaly correlation (ACC) versus lead week for six S2S systems. Top row: JFM 2026 (winter); bottom row: JJAS 2019 (monsoon). Columns: precipitation, Z500, 2-m temperature. Dotted line ACC= 0.5; solid line ACC= 0. DLESyM provides no precipitation; UKMO T2M and Spire (JJAS) are unavailable.

to 0.19. By week 4 both FuXi-S2S and DLESyM have crossed into *negative* ACC (−0.24 and −0.18) and remain negative through week 6 (−0.40 and −0.62)—meaning their week-4–6 Z500 forecasts are anticorrelated with the true anomaly pattern more often than not. Spire is the only system whose Z500 skill does not degrade monotonically with lead: its ACC is 0.69–0.77 across weeks 3–6, the highest of any individual system from week 4 onward, including the dynamical systems.

T2M. The T2M comparison is restricted to the three systems with a usable T2M product (Spire, FuXi-S2S, DLESyM; UKMO T2M is excluded per Section 3.5, and ECMWF/NCEP do not deliver a comparable T2M field in our archive). Spire again leads at every lead (Table **??**): ACC 0.75 at week 1 versus 0.49 for DLESyM and 0.33 for FuXi-S2S, and it remains the only system with ACC > 0.2 from week 3 onward. DLESyM's T2M skill follows the same long-lead pattern as its Z500 skill, turning negative by week 4.

5.3 Monsoon (JJAS 2019): the great equalizer

Precipitation. The winter-season separation between Spire and the rest disappears entirely in the monsoon, because the pattern that separated systems in JFM—one system clearly ahead of the others—is simply absent (Table **??**). At week 1, FuXi-S2S, UKMO, and ECMWF are statistically indistinguishable (ACC 0.39–0.40), with NCEP trailing at 0.29. By week 3 every system has fallen to ACC ≤ 0.06, and by week 4 every individual system and the MME are at or below ACC 0.02. From week 4 onward all systems—without exception—return *negative*

ACC, meaning the forecasts at these leads carry no usable information about the sign of the weekly rainfall anomaly over India. This is a qualitatively different outcome from JFM 2026, where four of six systems retained positive precipitation skill through week 6.

Z500. The same convergence is visible for Z500, though the skill horizon is somewhat longer than for precipitation (Table **??**). ECMWF is the most skillful and most persistent system at every lead (ACC 0.92 at week 1, remaining positive at 0.21–0.25 by weeks 5–6); UKMO is second-most persistent. DLESyM, the highest-resolution ML system, starts competitively (ACC 0.85 at week 1, second only to ECMWF) but collapses fastest of any system: it is the first to cross into negative territory, at week 4 (-0.12), and remains the most negative system through week 6 (-0.18). FuXi-S2S shows an irregular trajectory (0.91, 0.64, 0.29, 0.31, -0.11, 0.11) that does not improve on the dynamical systems at any lead.

T2M. DLESyM is the only system with a usable T2M product in JJAS 2019. Its skill declines from ACC 0.78 at week 1 to 0.41 by week 6—a smoother, more gradual decay than either its own Z500 trajectory or any system's precipitation trajectory in this season, consistent with the comparatively high autocorrelation of near-surface temperature anomalies relative to rainfall and circulation at subseasonal lead.

5.4 Divergence among the ML systems

The Z500 results above contain a finding that is easy to miss when comparing ML systems only to dynamical baselines: the three ML systems do not behave like a single class of model. At week 1 in JFM 2026, Spire, FuXi-S2S, and DLESyM are all skillful and mutually close (Z500 ACC 0.94, 0.91, and 0.92 respectively; Table **??**)—a week-1 leaderboard would rank the three ML systems as roughly interchangeable. By week 4 this is no longer true: Spire is at 0.73, while FuXi-S2S and DLESyM are at -0.24 and -0.18. The same divergence recurs in JJAS 2019, where DLESyM starts second only to ECMWF (week-1 ACC 0.85) but is the first system of either season to fall below the dynamical baselines (Table **??**).

The practical implication is that *long-lead stability is a separate property from headline (week-1) skill*, and the three ML systems in this benchmark do not share it. Spire is the only ML system whose Z500 forecasts remain informative beyond week 3 in either season; FuXi-S2S and DLESyM, despite being built on different architectures (transformer and diffusion-based, respectively) and trained on different data, share a qualitatively similar failure mode at long lead. We do not have access to the internals of either system and so cannot attribute this directly to a specific architectural or training choice, but the pattern is consistent with the broader concern that ML systems optimized for short-range accuracy can develop compounding error growth once integrated well beyond their primary training horizon (Section 6). The practical caution for anyone selecting an S2S system based on a short evaluation window is that a single week-1 score is not sufficient evidence of usable skill at weeks 4–6.

Table 2: JFM 2026 all-India precipitation CRPSS versus climatology by lead week. Positive values indicate probabilistic skill above climatology.

Model	W1	W2	W3	W4	W5	W6
Spire AI-S2S	**0.61**	0.40	0.25	0.19	0.18	0.17
FuXi-S2S	0.45	0.27	0.31	0.32	0.29	**0.26**
ECMWF	0.59	**0.51**	**0.40**	**0.35**	**0.30**	0.25
UKMO	0.45	0.27	0.18	0.25	0.23	0.19
NCEP	0.42	0.33	0.29	0.26	0.24	0.15

Table 3: JFM 2026 all-India Z500 CRPSS versus climatology by lead week.

Model	W1	W2	W3	W4	W5	W6
Spire AI-S2S	0.59	0.41	0.28	**0.28**	**0.30**	**0.29**
FuXi-S2S	0.71	0.04	-0.26	-0.27	-0.02	0.13
DLESyM	0.56	0.34	0.09	-0.05	0.07	0.10
ECMWF	**0.83**	**0.58**	**0.37**	0.23	0.23	0.23
UKMO	0.77	0.32	0.10	0.13	0.13	0.13
NCEP	0.26	-0.19	-0.18	-0.13	0.02	0.02

5.5 Probabilistic skill and calibration

The deterministic ranking of Section 5.2 does not simply carry over to probabilistic skill. For precipitation (Table 2), Spire has the highest CRPSS at week 1 (0.61, narrowly ahead of ECMWF's 0.59), but ECMWF overtakes it from week 2 onward and leads for the remainder of the forecast horizon (e.g. 0.51 versus 0.40 at week 2, 0.30 versus 0.18 at week 5). For Z500 (Table 3), ECMWF leads more clearly at short lead (CRPSS 0.83 at week 1 versus 0.59 for Spire) and remains ahead through week 3, but Spire becomes the most probabilistically skillful individual system from week 4 onward (0.28–0.30 versus ECMWF's 0.23), mirroring its deterministic Z500 advantage at the same leads. FuXi-S2S and NCEP show the least stable CRPSS trajectories: FuXi-S2S Z500 CRPSS falls to -0.26 at week 3 before partially recovering, and NCEP Z500 CRPSS is negative at weeks 2–4.

Figure 3 shows why ECMWF's probabilistic precipitation skill exceeds its deterministic ranking. Every system is under-dispersive (SSR < 1, indicating overconfidence) at week 1—Spire is closest to calibrated (SSR 0.63) while FuXi-S2S is the most overconfident (SSR 0.11). By week 3, ECMWF's ensemble has grown to near-ideal calibration for precipitation (SSR 0.70) while Spire has *over*-corrected into over-dispersion (SSR 1.43, indicating its spread now exceeds its error). For Z500 the same pattern is sharper still: Spire's spread–skill ratio exceeds 1.8 by week 4, meaning its ensemble spread is now substantially larger than its actual error—a sign of growing overdispersion that does not by itself harm CRPSS (an overdispersive ensemble can still be skillful) but indicates that Spire's stated uncertainty should not be taken as quantitatively calibrated at long lead. ECMWF's Z500 spread–skill ratio stays closest to one across the full lead range (0.92 to 1.46 and back to 0.94), consistent with its century-scale ensemble design and dedicated reforecast-based calibration.

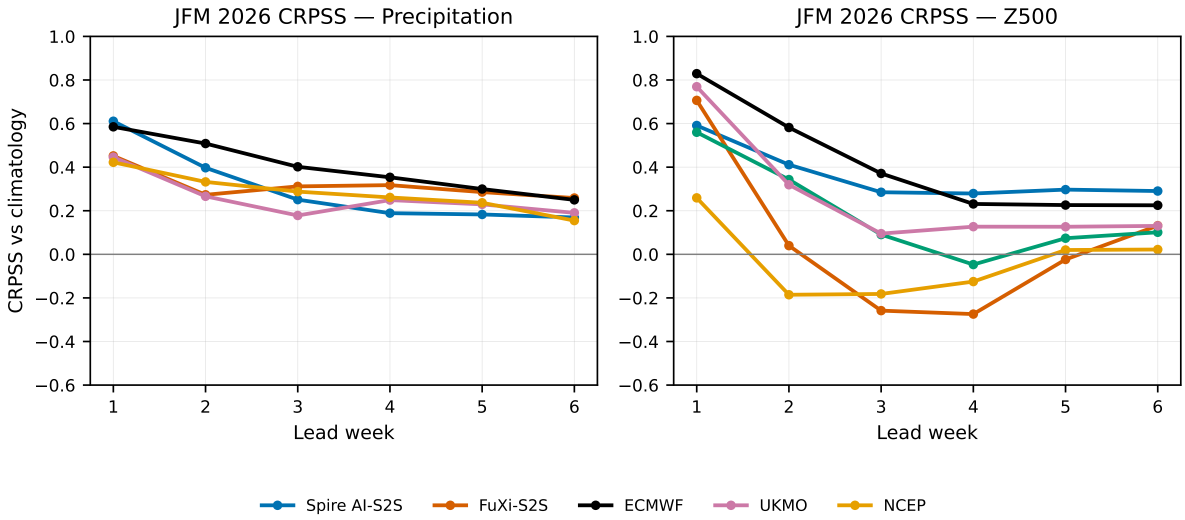

Figure 2: JFM 2026 all-India CRPSS versus lead week for precipitation (left) and Z500 (right). Positive values indicate probabilistic skill above climatology.

Table 4: JFM 2026 week-1 precipitation ACC by IMD homogeneous region. Best individual-model score in each column is bold.

Model	Northwest	Central	South Pen.	East/NE
Spire AI-S2S	**0.80**	0.75	**0.46**	**0.80**
FuXi-S2S	0.69	0.74	0.38	0.75
ECMWF	0.79	**0.79**	0.46	0.75
UKMO	0.75	0.71	0.38	0.75
NCEP	0.61	0.69	0.20	0.70
MME	0.78	0.77	0.37	0.76

5.6 Regional structure of skill

Figure 4 and Table 4 break down week-1 JFM 2026 precipitation ACC by the four IMD homogeneous regions (Section 2.1). Predictability is markedly unequal across India: Central India, East/Northeast India, and Northwest India are all well-predicted at week 1 by every system (ACC 0.61–0.80), while the South Peninsula is uniformly the hardest region (ACC 0.20–0.46 across all six systems)—roughly half the skill level of the other three regions for every model without exception. This regional gap is consistent with the synoptic character of winter precipitation in northern/central India versus the more locally forced, lower-amplitude rainfall the peninsula receives outside the northeast-monsoon season. Spire's domain-mean lead over the other systems (Section 5.2) is not concentrated in any one region: it is tied or within 0.01 of ECMWF in Central India (0.75 vs. 0.79) and Northwest India (0.80 vs. 0.79), and leads outright in East/Northeast India (0.80 vs. 0.75) and the South Peninsula (0.46, tied with ECMWF). No system—including Spire—shows a clear advantage specific to the hardest (South Peninsula) region, indicating that the winter ML advantage documented above is a broad-domain property rather than a regional artifact.

The same regional split is visible for Z500 (Table 5), though the gap between regions is much smaller than for precipitation: every system scores ACC 0.68–0.95 across all four regions at week 1, and even the hardest region (South Peninsula) remains highly skillful for every system

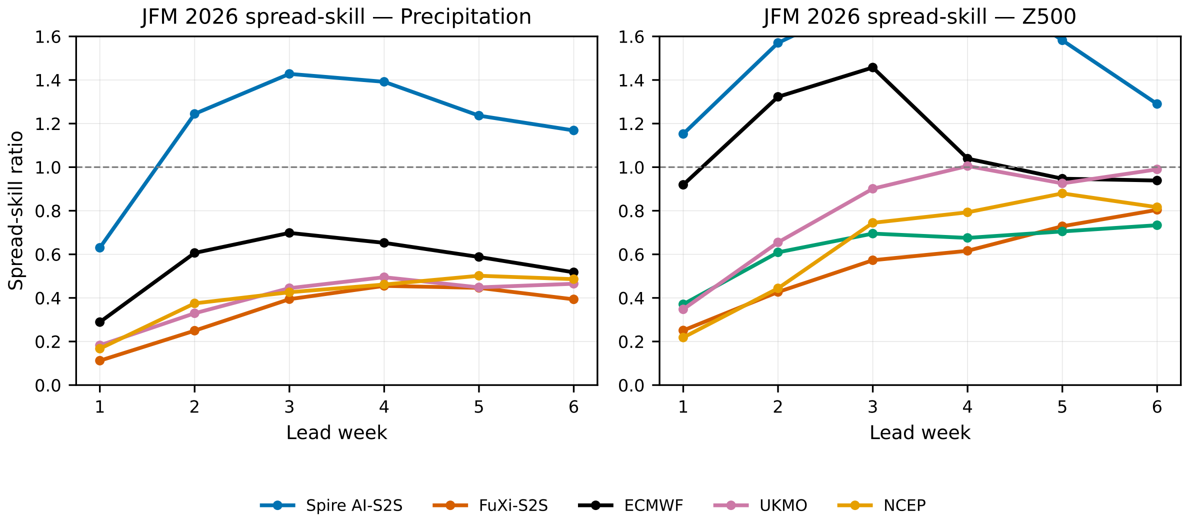

Figure 3: JFM 2026 spread–skill ratio versus lead week; a well-calibrated ensemble lies near unity (dashed line). Values < 1 indicate under-dispersion (overconfidence).

Table 5: JFM 2026 week-1 Z500 ACC by IMD homogeneous region.

Model	Northwest	Central	South Pen.	East/NE
Spire AI-S2S	0.91	0.91	0.79	0.87
FuXi-S2S	0.83	0.83	0.72	0.74
DLESyM	0.89	0.86	0.68	0.78
ECMWF	**0.95**	**0.94**	**0.89**	**0.93**
UKMO	0.92	0.93	0.87	0.78
NCEP	0.83	0.88	0.78	0.73
MME	0.94	0.93	0.86	0.88

(ACC 0.68–0.89). ECMWF leads in all four regions for Z500 at week 1, which is consistent with its domain-mean lead at this lead (Section 5.2); the much narrower spread between regions for Z500 than for precipitation reflects the larger spatial decorrelation scale of the 500-hPa height field relative to rainfall.

The monsoon precipitation collapse of Section 5.3 is also a broad-domain property rather than a regional artifact. At week 1 (Table 6) JJAS 2019 regional skill is already lower and flatter than JFM 2026: ACC ranges only 0.25–0.44 across all four regions and all four scored systems, versus the much wider 0.20–0.80 range in JFM (Table 4). By week 3 (Table 7) every region for every system has fallen to ACC ≤ 0.20, with most region–model combinations at or below zero; no region retains the kind of useful skill that Northwest, Central, and East/Northeast India still show in JFM 2026 at the same lead. East/Northeast India is a partial, model-dependent exception—ECMWF (0.20) and UKMO (0.16) retain some skill there at week 3 while collapsing elsewhere—but no system or region combination approaches the JFM 2026 winter skill level. The regional data therefore reinforce the domain-mean finding of Section 5.3: the monsoon equalizes skill across the whole of India, not just in the domain average.

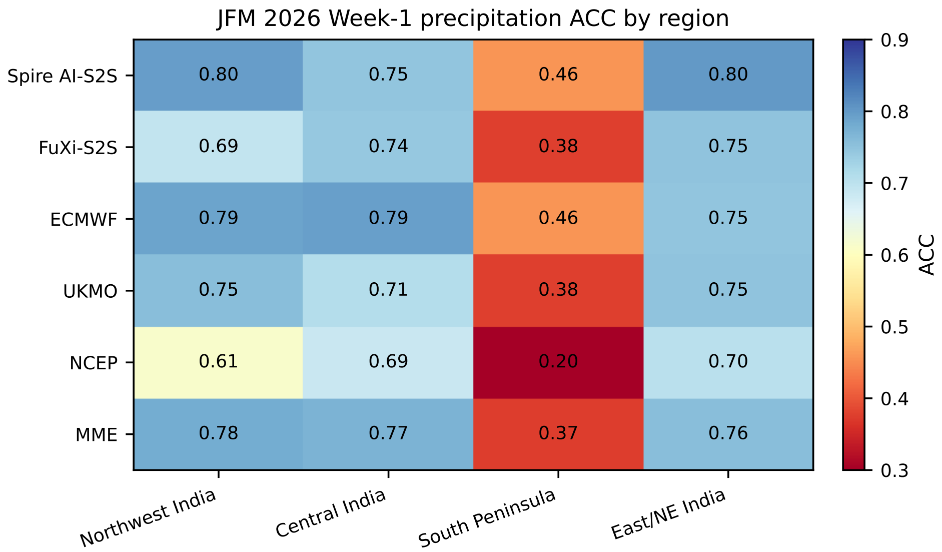

Figure 4: JFM 2026 week-1 precipitation ACC by IMD homogeneous region.

Table 6: JJAS 2019 week-1 precipitation ACC by IMD homogeneous region (IMD truth, 17 common initializations).

Model	Northwest	Central	South Pen.	East/NE
FuXi-S2S	0.40	0.33	0.31	0.33
ECMWF	**0.42**	0.33	0.32	**0.44**
UKMO	0.37	**0.36**	**0.34**	0.35
NCEP	0.29	0.28	0.25	0.01
MME	0.41	0.36	0.32	0.37

5.7 Robustness: larger initialization set

Because the 17-initialization common set (Section 4.4) is small, we repeat the JJAS 2019 precipitation verification on the independent 35-initialization operational subset (ECMWF, UKMO, NCEP; twice-weekly Monday/Thursday initializations, IMD truth) to check whether the monsoon collapse of Section 5.3 is an artifact of the smaller matched sample. It is not: week-1 ACC on the 35-initialization set is 0.39 (ECMWF), 0.39 (UKMO), and 0.28 (NCEP)—within 0.01–0.02 of the corresponding 17-initialization values—and skill falls to ACC ≤ 0.04 by week 3 and negative by week 4 for all three systems, exactly as in the matched-set result. The monsoon precipitation skill horizon of roughly two weeks is therefore a robust feature of JJAS 2019 across both initialization protocols, not a small-sample effect of the 17-date matched comparison.

6 Discussion

Where ML helps, and where it does not. The two-season design of this study lets us separate a genuinely better forecast system from one that merely benefits from an easy predictability regime. The answer is regime-dependent. In JFM 2026, winter precipitation over India is organized substantially by western disturbances—synoptic-scale systems with several days of coherent, predictable structure (Dimri et al., 2015; Hunt et al., 2018)—and Spire AI-

Table 7: JJAS 2019 week-3 precipitation ACC by IMD homogeneous region, showing the monsoon skill collapse is uniform across regions.

Model	Northwest	Central	South Pen.	East/NE
FuXi-S2S	0.04	-0.01	0.12	0.01
ECMWF	**0.13**	**0.04**	0.09	**0.20**
UKMO	0.00	-0.04	**0.14**	0.16
NCEP	0.01	-0.09	0.11	-0.12
MME	0.06	-0.03	0.12	0.12

S2S converts this structure into a real, lead-independent skill advantage over both operational dynamical systems and the other two ML systems (Section 5.2). In JJAS 2019, monsoon precipitation is dominated by smaller-scale convection embedded in a noisier large-scale envelope, and that advantage vanishes for every system, ML and dynamical alike (Section 5.3), confirmed on an independent initialization set (Section 5.7). The honest reading is not that ML systems are unskillful in the monsoon, but that *no system in this benchmark*, including the operational dynamical systems with decades of development behind them, retains useful weekly precipitation skill past week 2–3 in JJAS 2019. The winter result demonstrates that ML S2S forecasting can add real value over India when the underlying physics is favorable; the monsoon result demonstrates that this value does not yet generalize to the country's most consequential rainy season.

Not all AI systems are equal. A second contribution of this study is methodological: by evaluating three ML systems side by side rather than one ML system against dynamical baselines, we show that "ML skill" is not a single property. Spire is the only one of the three ML systems whose Z500 skill does not collapse to negative ACC within the six-week horizon, in either season (Section 5.4). FuXi-S2S and DLESyM, despite different architectures (transformer and diffusion-based respectively) and different training regimes, share a similar long-lead failure mode that is not visible from their week-1 scores alone. We cannot identify the specific architectural or training cause of this divergence from the forecast output alone, but it is consistent with compounding error growth once an autoregressive or iteratively sampled model is pushed well beyond the lead range it was primarily optimized for. The practical consequence for anyone selecting or comparing ML S2S systems is that a single short-lead evaluation window is not sufficient evidence of skill at the leads that matter for subseasonal decision-making.

Deterministic versus probabilistic skill. The third finding is that ranking systems by ensemble-mean ACC and ranking them by CRPSS can disagree. ECMWF leads precipitation CRPSS from week 2 onward despite Spire's higher ACC at every lead (Section 5.5), because ECMWF's 100-member ensemble, built on a mature operational reforecast-based calibration pipeline, achieves a spread–skill ratio closer to one across the full lead range, while Spire's delivered uncertainty becomes increasingly overdispersive beyond week 2. A forecast system that is the best deterministic choice is therefore not automatically the best choice for probabilistic, risk-based decision-making (e.g. threshold-exceedance warnings), and operational guidance for India should evaluate candidate S2S systems on both axes rather than either alone.

Consistency with prior S2S literature. Multi-model S2S assessments over South Asia report useful deterministic precipitation skill confined to roughly weeks 1–2, with regime- and teleconnection-dependent extensions at longer leads (de Andrade et al., 2019; Pegion et al., 2019; Mariotti et al., 2020). Our results are consistent with this picture in both seasons: JJAS 2019 precipitation skill is effectively exhausted by week 2–3 for every system, and even the most skillful JFM 2026 system (Spire) falls below ACC 0.5 by week 3. The Z500 skill horizon is correspondingly longer than the precipitation horizon in both seasons, consistent with the well-established result that large-scale circulation is predictable further into the subseasonal range than the surface-sensible weather it forces (Vitart et al., 2017; Rasp et al., 2024).

7 Limitations

We list the limitations of this study candidly, as several materially bound how the results should be interpreted.

Two seasons, not a multi-year hindcast. JFM 2026 and JJAS 2019 are single, specific seasons rather than a multi-year hindcast archive. The two-season contrast isolates a real regime difference (Section 6), but neither season's result should be read as a long-run climatological skill estimate for that season type; both winter and monsoon predictability vary year to year with the state of ENSO and other low-frequency drivers (Mariotti et al., 2020; Domeisen et al., 2022), which this single-year comparison cannot sample.

Different precipitation truth in each season. JFM 2026 precipitation is verified against ERA5 daily totals because the IMD gridded product for 2026 is not yet available; JJAS 2019 precipitation is verified against IMD (Section 2.2). This is the principled choice given current data availability, but it means the *within-season* model rankings reported here are robust while the *absolute* precipitation skill level should not be compared directly across the two seasons, since ERA5 precipitation has its own documented regional biases relative to gauge-based observations (Hersbach et al., 2020).

Spire is absent from JJAS 2019. Spire AI-S2S forecasts were not available for the 2019 season, so the central winter finding—that an ML system can lead an operational benchmark over India—cannot itself be tested against the monsoon. We can show that the dynamical-versus-ML gap closes for the five systems common to both seasons, but we cannot directly test whether Spire specifically would retain a monsoon advantage.

Common climatology, no per-model bias correction. All anomalies use a single climatology per truth source (Section 4.2) rather than each system's own reforecast-derived climatology, so lead-dependent model drift is not removed before scoring. This is the standard approach for a cross-system comparison without access to every system's reforecast archive, but it means some of the long-lead skill loss documented in Section 5 may reflect uncorrected mean-state drift rather than loss of anomaly information per se.

Unequal ensemble sizes. Ensemble size ranges from 4 (UKMO) to 100 (ECMWF) members across the dynamical systems, and Spire delivers a mean-and-spread product rather than individual members. UKMO's small ensemble in particular limits the precision of its probabilistic scores (Section 5.5) independent of the underlying forecast quality.

Variable coverage gaps. UKMO 2-m temperature is excluded throughout due to a data-retrieval issue at the source (Section 3.5); DLESyM does not provide a precipitation channel (Section 3.3). Neither gap is a property of the underlying forecast system, but both reduce the number of systems available for the T2M and precipitation comparisons respectively.

FuXi-S2S T2M bias. FuXi-S2S 2-m temperature carries an approximately 4 K cold bias relative to ERA5 at all leads, consistent with a mismatch between its 00 UTC initialization-time snapshot and the daily-mean reference used for verification. We report FuXi-S2S T2M scores as computed but flag this known offset; a systematic reference-time correction is a direct target for future work rather than a correction applied here.

8 Conclusions

We presented a two-season subseasonal benchmark over India, evaluating six S2S forecast systems—three machine-learning (Spire AI-S2S, FuXi-S2S, DLESyM) and three operational dynamical (ECMWF, UKMO, NCEP)—for precipitation, Z500, and T2M across the synoptically driven winter (JFM 2026) and the convectively dominated summer monsoon (JJAS 2019). The main conclusions are as follows.

(1) Spire AI-S2S leads winter forecasting over India across every variable and lead. It attains the highest precipitation ACC and lowest RMSE at every lead week (week-1 ACC 0.78 versus 0.73 for ECMWF/UKMO), leads T2M throughout, and is the only system whose Z500 skill does not collapse with lead, remaining the most skillful individual system from week 4 onward.

(2) The winter ML advantage does not survive the monsoon. Evaluated on a matched set of common initializations and confirmed on an independent 35-initialization robustness check, every system—ML and dynamical alike—loses usable weekly precipitation skill by week 3 in JJAS 2019, with no system retaining a meaningful edge over the others.

(3) The three ML systems are not interchangeable. Spire, FuXi-S2S, and DLESyM look similar at week 1 but diverge sharply by week 3–4: FuXi-S2S and DLESyM both fall to negative Z500 ACC in both seasons, while Spire does not in either. Long-lead stability is therefore a property that must be evaluated directly and cannot be inferred from short-lead skill alone.

(4) Deterministic and probabilistic rankings disagree. ECMWF leads precipitation CRPSS from week 2 onward despite Spire's higher ACC throughout, reflecting ECMWF's better-calibrated ensemble spread; a system favored by deterministic verification is not automatically the better choice for probabilistic, risk-based use.

Within the limits of two single seasons (Section 7), these results give an early, honest picture of where ML subseasonal forecasting adds value over India—the synoptically organized winter—and where the problem remains genuinely hard for every system evaluated, ML and operational alike—the summer monsoon. The verification framework, code, and result tables are archived to support extension as additional seasons, an IMD-2026 precipitation truth, and per-model reforecast climatologies become available.

Data and Code Availability

ERA5 reanalysis is available from the Copernicus Climate Data Store (`https://cds.climate.copernicus.eu`; Hersbach et al. 2020); the Analysis-Ready, Cloud-Optimized subset used here follows Carver and Merose (2023). IMD gridded daily rainfall follows Pai et al. (2014) and is distributed by the India Meteorological Department. ECMWF extended-range, UKMO GloSea6, and NCEP CFSv2 forecasts are available via the ECMWF Copernicus Data Store under the terms of the S2S database (Vitart et al., 2017). FuXi-S2S model weights and inference code are publicly available on Zenodo and GitHub respectively. Spire AI-S2S forecasts were provided under a research data-use agreement with Spire Global. DLESyM forecasts were obtained from the model providers (Cresswell-Clay et al., 2025). All verification, analysis, and figure-generation code for this study, including the scripts that regenerate every table and figure in this paper directly from the underlying result files, is available at `https://github.com/Artamta/s2s-analysis`.

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